

Solution to Wrap Around Relay

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1 Solution

Let's first write each variable in terms of the other. Question 1 is pretty straightforward, the expression is already in terms of A. However, we can complete the square to simplify the expression a little more.

$$B = A^2 - 40A + 388 = (A - 20)^2 - 12$$

Question 2 is a combo problem. Each person wants to sit with their friend, so we can just treat them as $\frac{B}{2}$ pairs with $\frac{B}{2}$ seats. To account for rotations, we can fix one pair at a certain position and then choose the places for the $\frac{B}{2} - 1$ remaining pairs. Thus there are $(\frac{B}{2} - 1)!$ ways to seat them. However, within each pair, we could have person 1 then person 2, or person 2 then person 1. So we have 2 choices for each pair, or a multiplier of $2^{\frac{B}{2}}$.

$$C = (\frac{B}{2} - 1)! 2^{\frac{B}{2}}$$

Question 3 tells us that A is however many multiples of C are less than 100.

$$A = \lfloor \frac{99}{C} \rfloor$$

Lets look at question 2 first, as its the most restrictive. From the problem statement, we know that B must be a positive even integer. Now lets trace that back to question 1. From our completing the square, B is positive when $(A - 20)^2 > 12$. Question 3 tells us that A is a positive integer, so we can conclude that: $A > 20 + \sqrt{12}$ meaning $A > 23$. More than that for B to be an even number, $(A - 20)^2$ must have the same parity as 12, so A has to be even as well. This means A can be 24, 26, 28, ...

Looking at the expression for A: $A = \lfloor \frac{99}{C} \rfloor$ tells us that C could be at most 4, as 5 or any number greater would give $A < 20$.

All that's left is to try out different values of C. The only value of C that produces an even A is 4. Plugging in the numbers you would get that $A = \lfloor \frac{99}{4} \rfloor = 24$, $B = (24 - 20)^2 - 12 = 4$, and $C = (\frac{4}{2} - 1)! 2^{\frac{4}{2}} = 1! \cdot 4 = 4$, which is what we set C to be. This confirms that $C = 4$ works as a solution and gets you that $(A, B, C) = (24, 4, 4)$.